

to basic neural network concepts like the artificial neuron, neuron layer, neuron connections, weight, transfer function, input function, learning rule, etc. Neuroph supports common neural network architectures such as multi-layer perceptron with Backpropagation, Kohonen and Hopfield networks. All these classes can be extended and customised to create custom neural networks and



learning rules. Neuroph has built-in support for image recognition.

Useful links

Neuroph home: <http://neuroph.sourceforge.net/>

GitHub: <https://github.com/neuroph/neuroph>

8) Deeplearning4j: The first commercial-grade, open source, distributed deep learning library written for Java and Scala

Deeplearning4j (DL4J) is integrated with Hadoop and Spark. DL4J is designed to be used in business environments on distributed GPUs and CPUs. Skymind is its commercial support arm, bundling Deeplearning4j and other libraries



such as TensorFlow and Keras in the Skymind Intelligence Layer (SKIL, Community Edition), which is a deep learning environment that gives developers an easy, fast way to train and deploy AI models. SKIL acts as a bridge between Python data science environments and the JVM.

Advantages

- Deeplearning4j aims to be cutting-edge plug-and-play, with more convention than configuration, which allows for fast prototyping for non-researchers.
- It is customisable at scale.
- DL4J can import neural net models from most major frameworks via Keras, including TensorFlow, Caffe and Theano, bridging the gap between the Python ecosystem and the JVM with a cross-team toolkit for data scientists, data engineers and DevOps. Keras is employed as Deeplearning4j's Python API.
- Machine learning models are served in production with Skymind's model server.

Features

- Distributed CPUs and GPUs
- Java, Scala and Python APIs
- Adapted for micro-service architecture
- Parallel training via iterative reduce
- Scalable on Hadoop
- GPU support for scaling on AWS

Libraries

- Deeplearning4j: A neural Net platform
- ND4J: Numpy for the JVM
- DataVec: A tool for machine learning ETL operations
- JavaCPP: The bridge between Java and native C++
- Arbiter: An evaluation tool for machine learning algorithms
- RL4J: Deep reinforcement learning for the JVM

Useful links

DL4J home: <https://deeplearning4j.org/>

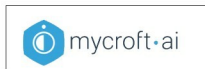
GitHub: <https://github.com/deeplearning4j/deeplearning4j>

SKIL: <https://skymind.ai/quickstart>

9) Mycroft: One of the world's first open source assistants, ideal for anything from a science project to an enterprise software application

Mycroft runs anywhere – on a desktop

computer, inside an automobile, or on a Raspberry Pi. This is open source



software which can be freely remixed, extended and improved. Mycroft may be used in anything from a science project to an enterprise software application.

Useful links

MyCroft home: <https://mycroft.ai>

GitHub: <https://github.com/MycroftAI/mycroft-core>

10) OpenCog: A project that aims to build an open source artificial intelligence framework

OpenCog is a diverse assemblage of cognitive algorithms, each embodying their own innovations. But what makes the overall architecture powerful is its careful adherence to the principle of cognitive synergy. OpenCog was originally based on the 2008 release of the source code of the proprietary Novamente Cognition Engine (NCE) of Novamente LLC. The original NCE code is discussed in the PLN book (reference given below). Ongoing development of OpenCog is supported by the Artificial General Intelligence Research Institute (AGIRI), the Google Summer of Code project, and others.



OpenCog Prime is the architecture for robot and virtual embodied cognition, which defines a set of interacting components designed to give rise to human-equivalent artificial general intelligence (AGI) as an emergent phenomenon of the whole system. OpenCog Prime's design is primarily the work of Ben Goertzel, while the OpenCog framework is intended as a generic framework for broad-based AGI research.